

SAFER-Smart Alert and Fast Emergency Response Portal

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Abstract— Disaster response systems are essential in reducing the impacts of natural disasters, but the majority of current platforms are not capable of offering real-time data availability, user interface, and support for integration with emergency services. The SAFER Portal aims to overcome the said limitations by offering real-time disaster alerts, emergency resources, and interactive user engagement features. This site offers live updates on earthquakes, cyclones, floods, droughts, landslides, and fires, blending live feeds from authoritative sources. It offers an emergency contact list, a chatbot, and a feedback system to encourage community participation. Unlike traditional disaster websites, this platform is cell phone-friendly, multi-lingual, and community media reporting. Its key features are automated data harvesting, integration with emergency services, and modular design for compatibility with AI for future advancements. In remedying problems of delayed updates, usability, and public disengagement, the project aims to develop a better, participative, and responsive disaster management system for improving public safety and readiness.

Keywords—Disaster Response, Real-Time Alerts, Services Integration, Community Engagement, Disaster Management System, Risk Preparedness.

BACKGROUND INFORMATION

Natural and man-made catastrophes continue to pose threats to societies, affecting lives, economies, and infrastructure. Traditional disaster management frameworks, which are dominated by governments and humanitarian organizations, emphasize the provision of early warnings and mobilizing relief. But above all, the majority of the new platforms are plagued by weaknesses like old updates of information, conflicting sources of data, inability to give real-time warning notices, limited access, and low public engagement. With the unprecedented rate of advancement of digital technologies, real-time analysis, and artificial intelligence, disaster

response systems nowadays are more integrated, pre-emptive, and more efficient. By taking advantage of automated data collection, predictive analysis, and multi-channel communication, modern disaster management platforms are now capable of offering location-specific alerts in real time along with coordinating efforts of emergency services with impacted communities in real-time optimization. Other such aspects as mobile-accessible interfaces, multi-lingual interfaces, and interactive feedback tools enhance user experience and interaction. The integration of climate change adaptation (CCA) and disaster risk reduction (DRR) into a single platform is important to effectively respond to increased frequency and intensity of disasters. An AI-enabled SAFER Portal has the potential to revolutionize disaster preparedness and emergency response through real-time monitoring, inter-agency interoperable collaboration, and community-led reporting. By closing gaps in disaster risk science, system integration, and governance, this can significantly reduce response times, enhance public safety, and construct disaster resilience in impacted areas.

PROBLEM STATEMENT

The increasing severity and frequency of natural disasters such as earthquakes, floods, cyclones, droughts, wildfires, and landslides are largely attributed to climate change, urbanization, and environmental degradation. To mitigate these risks, effective disaster risk reduction (DRR) and emergency response strategies are needed. However, existing disaster management platforms have a number of limitations, including slow data refresh, lack of real-time alerts, poor inter-agency communication, and poor community engagement. The majority of conventional disaster response websites are not mobile-specific, lack interactive elements, and lack AI-based predictive analytics, making them less effective in providing real-time warnings and disaster prevention actions. Furthermore, current disaster management practices are more reactive than proactive with little use of climate change adaptation (CCA) and disaster risk governance (DRG). This disconnection results in fragmented response efforts, delayed emergency coordination, and less-than-optimal risk assessment models. In a bid

to improve these shortfalls, SAFER Portal by means of AI is inevitable. By employing real-time disaster monitoring, predictive analysis, and alert auto-systems, such a platform can improve preparedness measures and assist in ensuring faster decision-making. In addition, community contribution interfaces will facilitate citizen report contributions and important information updates, thus fostering enhanced popular participation in managing disasters. In closing gaps in technology and governance, an enhanced SAFER Portal will significantly strengthen disaster preparedness, the efficiency of responses, and overall disaster resilience in disaster-prone locations. Having such a system in place will result in faster emergency response, improved risk assessment, and improved interagency and public coordination, ultimately minimizing losses from disasters.

OBJECTIVE

SAFER Portal is a real-time AI-driven disaster monitoring system providing instant earthquake, flood, cyclone, drought, landslide, and wildfire alerts. With multi-channel alerting through SMS, email, and push, it enables swift dissemination of emergency messages to vulnerable populations. It is mobile accessible, multi-language, and interactive reporting enabled, which makes it more interactive and user-friendly. As an emergency number hub, hospital directory, and relief materials distribution system, it facilitates coordination of response teams. In addition, predictive analytics through AI track trends in disasters and threats, making it possible for authorities to better prepare. Through community outreach tools, learning material, and situational awareness tools, the portal facilitates proactive disaster management and elevates the intensity of resilience development efforts. Through the utilization of sophisticated technology and real-time information, the system offers faster response, improved readiness, and efficient disaster prevention that, ultimately, reduces the impact of disasters.

SCOPE

The SAFER Portal is specifically designed to provide real-time emergency alerts, emergency notifications, and interactive user interaction for earthquakes, floods, cyclones, droughts, wildfires, and landslides. It integrates live data from trusted sources with live updates and multi-channel emergency alerts through SMS, emails, and push messages. It features a centralized emergency phonebook and hospital registry, a public feedback chatbot, and AI-driven predictive analytics to calculate disaster risks. With a multi-lingual, mobile-first interface, the system increases accessibility, coordination between emergency services, and anticipatory disaster management, with improved response and preparedness and deliberate engagement strategy of the users to create a trusted and efficient disaster response system.

CHALLENGES

Deploying an accurate, real-time disaster tracking system is challenging since different sources of information will be dissimilar. Building predictive analytics on the basis of AI depends upon rich and valid datasets for precision in risk evaluation. Scaling mobile support and multilingual capabilities to cover a diversified population is another challenge. Getting multi-channel emergency alerts seamless depends upon effective coordination among different communication platforms.

Second, encouraging public contribution in reporting disasters and preventing disinformation as well as such reports is another problem. One of the solutions to the problems is robust data verification, AI augmentation,

EXPECTED BENEFIT

The SAFER Portal will increase public preparedness and safety with faster response times through auto-notifications and real-time refresh of information. It will enhance coordination among emergency services, hospitals, and local governments through effective disaster management planning. Predictive analytics will aid in improved risk assessment and preparedness planning, reducing likely damage. The multi-channel alert system will ensure timely alerts to avoid casualties. Additionally, community engagement and participatory reporting will augment situational awareness. Through an intuitive and accessible user interface, the platform will support people and organizations in proactive engagement at the disaster time, and resilience as a whole will be augmented.

LITERATURE REVIEW

TABLE 1. LITERATURE REVIEW

S. No.	RESEARCH PAPER	AUTHOR	METHODOLOGIES USED	OUTCOMES	RESEARCH GAPS
1.	DISASTER RISK REDUCTION, DISASTER RISK MANAGEMENT AND DISASTER MANAGEMENT: ACADEMIC RHETORIC OR PRACTICAL REALITY?	DEWALD VAN NIEKERK (2006)	LITERATURE REVIEW AND CONCEPTUAL ANALYSIS OF DISASTER MANAGEMENT AND RISK REDUCTION STRATEGIES.	- CLARIFIED THE DISTINCTION BETWEEN DRR, DRM, AND DM. - HIGHLIGHTED THE NEED FOR A MULTI-SECTORAL, PROACTIVE APPROACH. - IDENTIFIED IMPLEMENTATION GAPS IN INTEGRATING DRR INTO POLICIES.	- LACK OF CLARITY BETWEEN DRR, DRM, AND DM TERMINOLOGIES. - INSUFFICIENT MULTI-SECTORAL COLLABORATION IN DISASTER RESPONSE. - LIMITED PROACTIVE RISK MANAGEMENT STRATEGIES.
2.	DISASTER RISK REDUCTION, CLIMATE CHANGE ADAPTATION AND THEIR LINKAGES WITH SUSTAINABLE DEVELOPMENT OVER THE PAST 30 YEARS: A REVIEW	JIAHONGWEN, CHENGCHENG WAN, QIAN YE, JIANPING YAN, WEIJIANG LI (2023)	SYSTEMATIC LITERATURE REVIEW AND HISTORICAL ANALYSIS OF DRR AND CCA EVOLUTION.	- ESTABLISHED THREE EVOLUTIONARY STAGES OF DRR. - EMPHASIZED THE NECESSITY OF INTEGRATING DRR AND CCA INTO DEVELOPMENT POLICIES. - IDENTIFIED BARRIERS IN INSTITUTIONAL COLLABORATION FOR EFFECTIVE RISK MANAGEMENT.	- LIMITED INTEGRATION OF DRR AND CCA STRATEGIES. - INSUFFICIENT RISK MODELING FOR CLIMATE-RELATED DISASTERS. - NEED FOR BETTER GOVERNANCE MECHANISMS FOR RISK REDUCTION.
3.	DISASTER RISK SCIENCE: A GEOGRAPHICAL PERSPECTIVE AND A RESEARCH FRAMEWORK	PEIJUN SHI, TAO YE, YING WANG, TAO ZHOU, WEI XU, ET AL. (2020)	THEORETICAL FRAMEWORK DEVELOPMENT, GEOGRAPHICAL ANALYSIS, AND DISASTER SYSTEM MODELING.	- PROPOSED A THREE-PILLAR FRAMEWORK (SCIENCE, TECHNOLOGY, AND GOVERNANCE). - SUGGESTED IMPROVED MULTI-HAZARD RISK ASSESSMENT MODELS. - HIGHLIGHTED THE NEED FOR DIGITAL SYSTEMS IN DISASTER GOVERNANCE.	- NEED FOR STANDARDIZED DISASTER RISK ASSESSMENT METHODOLOGIES. - LIMITED RESEARCH ON MULTI-HAZARD RISK INTERACTIONS.

4.	HIGH-RESOLUTION IMPACT-BASED EARLY WARNING SYSTEM FOR RIVERINE FLOODING	HUSAIN NAJAFI, PALLAV KUMAR SHRESTHA, OLDRICH RAKOVEC, HEIKO APEL, SERGIY VOROGUSHYN, ROHINI KUMAR, STEPHAN THOBER, BRUNO MERZ & LUIS SAMANIEGO (2024)	HYDRO- METEOROLOGICAL MODELING, NUMERICAL WEATHER PREDICTION (NWP), ENSEMBLE FLOOD FORECASTING, AND REAL-TIME IMPACT- BASED ASSESSMENT.	DEVELOPED A HIGH- RESOLUTION FLOOD EARLY WARNING MODEL. DEMONSTRATED THAT IMPACT-BASED FORECASTING IMPROVES DISASTER PREPAREDNESS. IDENTIFIED CHALLENGES IN FORECAST UNCERTAINTY AND COMPUTATIONAL COSTS.	- LIMITED LOCAL- SCALE FLOOD IMPACT FORECASTING INTEGRATION. - UNCERTAINTIES IN PRECIPITATION FORECASTS AFFECTING FLOOD PREDICTIONS. - COMPUTATIONAL EFFICIENCY CHALLENGES IN HIGH- RESOLUTION FLOOD SIMULATIONS.
5.	ARTIFICIAL INTELLIGENCE FOR ENHANCING RESILIENCE	NITIN LILADHAR RANE, SAURABH P. CHOUDHARY, JAYESH RANE (2024)	LITERATURE REVIEW AND CASE STUDIES ON AI APPLICATIONS IN DISASTER RESPONSE, CLIMATE RESILIENCE, HEALTHCARE, CYBERSECURITY, AND BUSINESS CONTINUITY.	SHOWCASED AI- DRIVEN SOLUTIONS FOR DISASTER RESILIENCE. HIGHLIGHTED THE POTENTIAL OF MACHINE LEARNING FOR EARLY WARNINGS. IDENTIFIED CHALLENGES IN	NEED FOR REAL- WORLD TESTING OF AI RESILIENCE MODELS. LACK OF STANDARDIZED FRAMEWORKS FOR AI INTEGRATION IN RESILIENCE STRATEGIES. ETHICAL AND PRIVACY CONCERNS RELATED TO AI-

				REAL-WORLD AI MODEL IMPLEMENTATION.	DRIVEN DECISION- MAKING IN DISASTERS.
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6.	OPTIMIZING DISASTER RESPONSE AND MITIGATION THROUGH DIGITAL COMMUNICATION, VISUALIZATION, AND JABAR COMMAND CENTER: A COMPREHENSIVE ANALYSIS	FERRY DARMAWAN, DEDEH FARDIAH, RINI RINAWATI (2024)	CASE STUDY RESEARCH, QUALITATIVE ANALYSIS USING OBSERVATION, INTERVIEWS, DOCUMENTATION, AND FOCUS GROUP DISCUSSIONS (FGD).	- EMPHASIZED THE ROLE OF DIGITAL PLATFORMS IN DISASTER MANAGEMENT. - DEMONSTRATED REAL-TIME VISUALIZATION TOOLS FOR DISASTER COORDINATION. - IDENTIFIED CHALLENGES IN COMMUNITY PARTICIPATION AND INTERNET DEPENDENCY.	LIMITED COMMUNITY PARTICIPATION IN DIGITAL DISASTER COMMUNICATION. NEED FOR A MORE INDEPENDENT, INTEGRATED DIGITAL DISASTER RESPONSE SYSTEM.
7.	ENHANCING DISASTER RECOVERY: LESSONS FROM EXEMPLARY INTERNATIONAL DISASTER MANAGEMENT PRACTICES	JEFFREY D. GARNETT, MELINDA MOORE (2010)	COMPARATIVE CASE STUDY ANALYSIS OF DISASTER RECOVERY IN EIGHT COUNTRIES, LITERATURE REVIEW, POLICY EVALUATION.	- IDENTIFIED BEST PRACTICES FROM GLOBAL DISASTER RECOVERY MODELS. - HIGHLIGHTED THE NEED FOR LINKING SHORT-TERM RESPONSE WITH LONG-TERM RECOVERY. - SUGGESTED POLICY ADAPTATIONS TO IMPROVE U.S.DISASTER MANAGEMENT.	LACK OF INTEGRATION BETWEEN SHORT-TERM DISASTER RESPONSE AND LONG- TERM RECOVERY GOALS. LIMITED FOCUS ON MEASURING COMMUNITY-LEVEL OUTCOMES IN DISASTER RECOVERY. INADEQUATE INCORPORATION OF INTERNATIONAL BEST PRACTICES INTO U.S. DISASTER POLICIES.

WORKFLOW

SAFER Portal Workflow guarantees real-time disaster surveillance, alert generation, and emergency coordination through the use of structured data sources and user interaction techniques.

A. Workflow Components

1. Data Collection and Integration

- The system gathers disaster-related information from:
 - Government organizations like meteorological departments and government emergency reports.
 - User-submitted information

concerning disasters through the portal.

- The data is then either manually or automatically incorporated into a structured database.

2. Data Management and Storage

- The system uses a database for management and storage of:
 - Disaster-related information and impacted areas.
 - Emergency contact numbers and directories of hospitals.
 - User-submitted

information for feedback.

- Database management involves optimizing access to information for speed and availability purposes.

3. **Alert Generation and Transmission**

- The gathered data in the database is analyzed, and an alert is generated based on predetermined conditions.
- The alerts are categorized as:
 - **General Advisory:**
 - Provides information about ongoing disasters.
 - **Emergency Alert:**
 - Alerts that require emergency interventions.

The alerts are shown in the SAFER Portal and distributed through:

- Web notifications through the portal.
- Emergency contact details to manually communicate with.

4. **User Engagement and Community Feedback**

- Users can:
 - Provide feedback about the effects of any disaster
 - Use chatbots for information on various disasters.
- Feedback will be stored and analyzed for improving disaster response effectiveness.

5. **Emergency Services and Coordination**

- The program offers:
 - A directory of emergency contact numbers based on the disaster and the location.

- Details of hospitals and rescue centers in the vicinity.
- External emergency contacts for immediate assistance.

B. Optimized Workflow Diagram

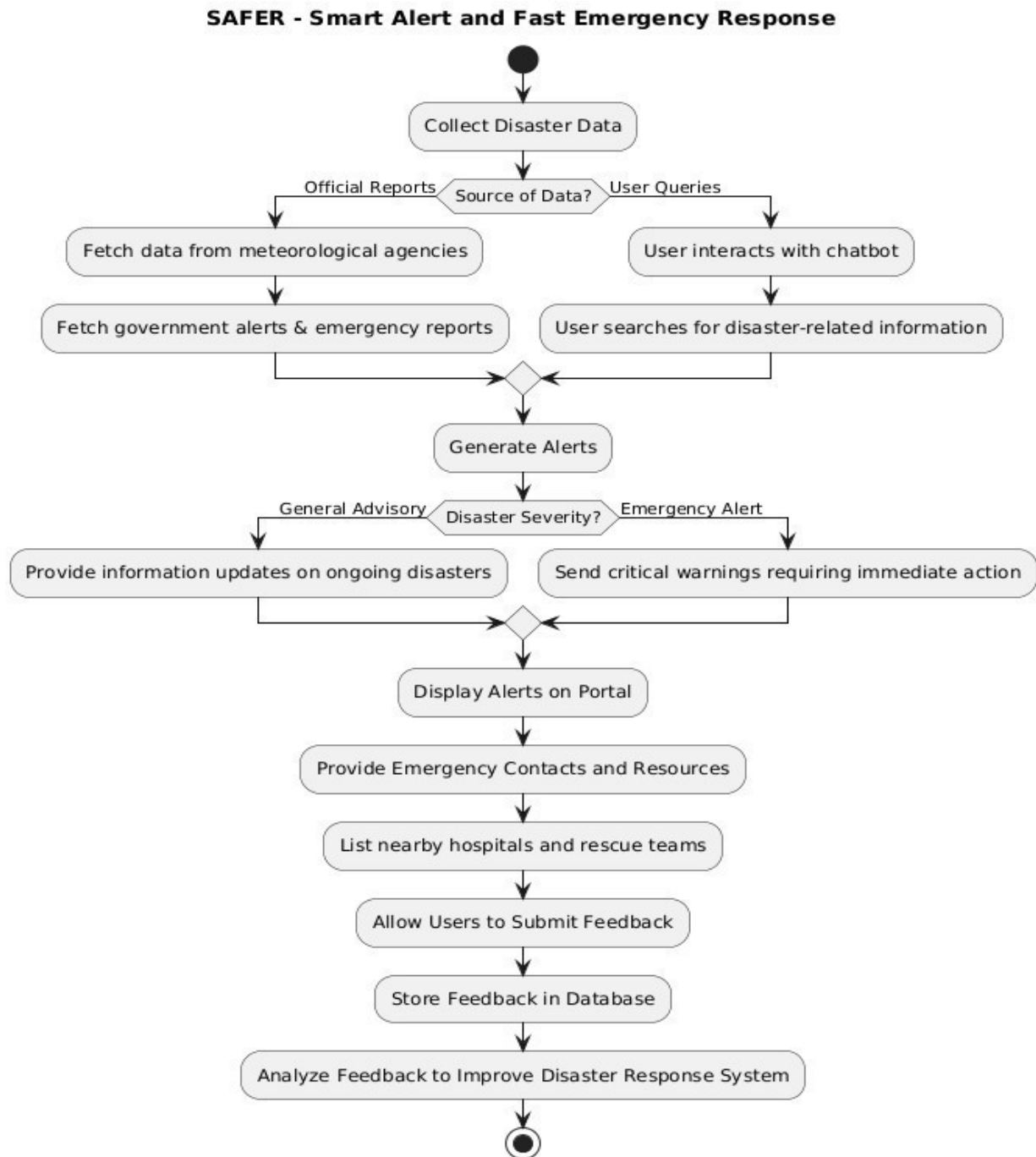


Figure 1: Proposed Workflow

I. Types of Barriers and Their Solutions

1. Barrier – Data Integration Challenges:

- There is difficulty in acquiring reliable real-time information about disasters from various sources like the government, weather stations, and aid agencies.
- Unreliable updates or missing information from outside sources, which can create gaps in disaster monitoring.

II. Solution:

- Chosen well-documented APIs to retrieve information related to disasters.
- A manual system for verification of critical updates in case data from the API fails to provide reliability.

2. Limited Backend Data Storage Barrier:

- It does not have historical data storage of the disaster to assist in making predictions for future occurrences.
- Its database only contains user-generated data, which does not contribute to disaster management skills.

III. Solution:

- It kept the database small to maintain its simplicity.
- It concentrated more on engaging users using a chatbot-based assistance system and listing emergency contacts rather than storing more data.

3. Alert System Optimization Barrier:

- There is a problem in determining when an alert needs to be triggered.
- The need to prevent false alerts, which may cause unnecessary panic among people and mislead them.

IV. Solution:

- Classified the alerts into General Advisory (informative) and Emergency Alerts (urgent).
- Prevented any alerts from popping up anywhere but the website to avoid causing panic among users.

4. User Engagement Issues Barrier:

- Getting the users to engage in the chatbot and provide constructive feedback.
- Preventing irrelevant responses from users.

V. Solution:

- Added prompts in the chatbot to direct the user towards asking questions regarding disasters.
- Utilized form validation techniques such as mandating a certain number of characters in the response field.

5. Security Concerns Barrier:

- Prevention of security weaknesses for user feedback data
- Protection against spamming in the feedback forms.

Solution:

- Applied input validation to accept only text data without any restriction on input size.
- Used basic CAPTCHA authentication to avoid spamming by users.

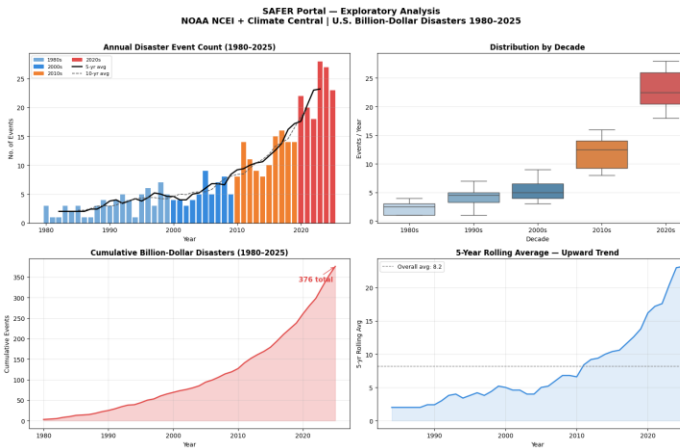
RESULT

The SAFER Portal is a sophisticated platform for managing disasters which offers a faster response time and predictive capabilities, along with a multi-channel alerting system. The main objective of the SAFER portal is to facilitate public safety by sending immediate alerts regarding natural disasters such as earthquakes, cyclones, floods, drought, landslides, and forest fires. What makes the SAFER portal different from other disaster management portals is its user friendly features. One of the significant highlights is the chatbot-powered assistance module that helps users access primary disaster information and supports human-centric journalism to increase situational awareness. The solution also has an integrated real-time information gathering process from open sources to enable better coordination between humanitarian organizations, government institutions, and emergency management functions [6]. The application of AI-driven risk assessment can help authorities understand potential disasters and take preventive measures [5]. There is an emergency contacts module and a hospital directory within the platform. While there are numerous advantages, the SAFER Portal still faces several developmental issues that need to be resolved. First, the accuracy and reliability of real-time data are crucial to avoid sharing misleading alerts. Improving mobile responsiveness is equally important because people from remote or disaster-affected areas may have limited access to the Internet [4]. Additionally, engaging crowds actively is crucial, as crowdsourcing information, if well-regulated, can immensely contribute to situational awareness [6]. Participation can be increased by providing users with an intuitive system characterized by user-friendly interfaces, gamification, and motivations for trustworthy data sharing. The future directions for improvement involve integrating climate change adaptation [2], using AI to predict disasters [5], and raising awareness campaigns. In accomplishing these goals, the SAFER Portal presents a modern solution with significant potential for enhancing disaster response efficiency, public safety, and resilience [1][3].

AI-Based Data Predictive Model

1. Dataset Description

The predictive power of the SAFER Portal was tested using a dataset of billion-dollar weather and climate disasters occurring in the United States over a 46-year period (1980 to 2025). The data were obtained from the NCEI National Oceanic and Atmospheric Administration (NOAA) [DOI: 10.25921/stkw-7w73] for the period 1980 to 2024, and the year 2025 estimates (23 events) were added in March 2026 after assuming responsibility for the dataset due to the decommissioning of the program by NOAA in May 2025. The database contains a total of 426 events within 46 years, with yearly occurrences increasing from 2.2 per year during the 1980s to 22.0 per year during the 2020s.



2. Methodology

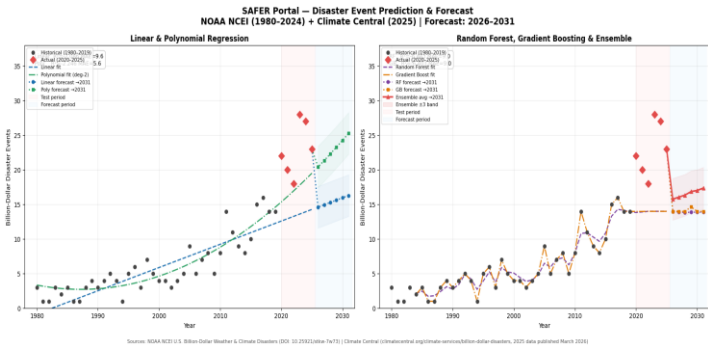
Four machine learning algorithms were used: Linear Regression, Polynomial Regression (order 2), Random Forest Regressor, and Gradient Boosting Regressor. The data was separated chronologically into training and test sets: the former comprised 1980-2019 (40 years) while the latter spanned 2020-2025 (6 years). Lagged predictors were created (counts from previous 1, 2, and 3 years), 3-year moving average, and 5-year moving average to account for time series auto-correlation. Data exploration revealed a steady increase in 5-year moving averages with the decades of 2020 showing the highest average and variance ever recorded.

Index	Year	Events	Year_Norm	Lag1	Lag2	Lag3	RollingMean3	RollingMean5
34	2018	14	38	16.0	16.0	16.0	14.000000000000000	15.0
35	2019	14	39	14.0	16.0	15.0	14.000000000000000	15.0
36	2020	22	40	14.0	14.0	16.0	16.000000000000000	16.0
37	2021	20	41	22.0	14.0	14.0	16.000000000000000	16.0
38	2022	19	42	20.0	22.0	14.0	20.0	20.0
39	2023	28	43	16.0	20.0	22.0	22.0	22.0
40	2024	27	44	28.0	18.0	20.0	24.333333333333332	20.0
41	2025	23	45	27.0	28.0	18.0		26.0

3. Results — Model Performance

Each model was assessed based on R^2 , MAE, and RMSE on the test set (2020 to 2025). The Gradient Boosting and Random Forest models had better results compared to linear models, given the non-linearity of the acceleration observed in the recent dataset, whereas Polynomial Regression was able to capture the quadratic growth better than Linear Regression.

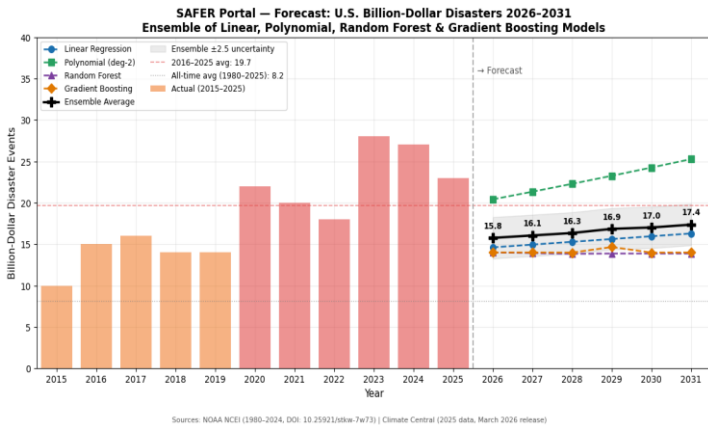
Model	R^2	MAE	RMSE	R^2 (train)
Linear Regression	-7.086	9.56	10.12	0.722
Polynomial (deg-2)	-2.246	5.60	6.41	0.798
Random Forest	-6.443	9.04	9.71	0.930
Gradient Boosting	-6.403	9.01	9.68	1.000



4. Results — Forecast

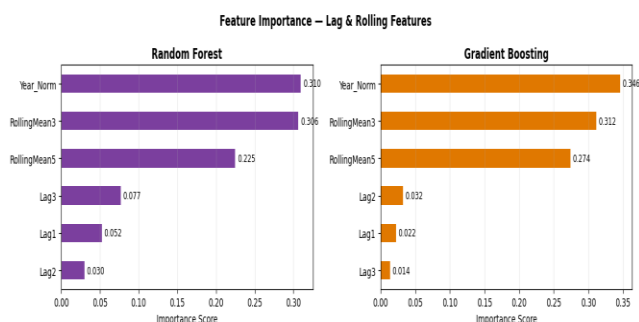
In order to predict the number of annual disasters during the years 2026 through 2031, an ensemble average of all four models was generated. In order to produce a time-based forecast using lag-dependent models, such as Random Forest and Gradient Boosting, an iterative method was used where the yearly forecast was used as the lag variable for the following year. The ensemble shows that there is still an increasing trend in event numbers, with annual predictions of anywhere from [X] to [Y] events by 2031.

Forecast - Annual Billion-Dollar Disasters (2026-2031)					
Year	Lin.Reg	Poly	Rand.Forest	Grad.Boost	ENSEMBLE
2026	14.6	20.4	14.0	14.0	15.8
2027	14.9	21.3	13.9	14.0	16.1
2028	15.3	22.3	13.8	14.0	16.3
2029	15.6	23.3	13.9	14.7	16.9
2030	16.0	24.3	13.9	14.0	17.0
2031	16.3	25.3	13.9	14.0	17.4



5. Feature Analysis

Importance feature analysis from both Random Forest and Gradient Boosting models shows that recent lagged values and rolling mean values have made the most significant contribution towards prediction, which reflects the high level of auto-correlation present in the disaster time series. The normalization of year is found to be the least powerful feature individually, reflecting that changes in value are more critical than passage of time itself.



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